# Mohammad Askari

Los Angeles, CA | askari@ucla.edu | LinkedIn | GitHub | Google Scholar

## EDUCATION

**UNIVERSITY OF CALIFORNIA, LOS ANGELES (UCLA)**

PhD Candidate in Civil Engineering — cGPA 4.0

MSc Candidate in Statistics — cGPA 3.9

## TECHNICAL EXPERIENCE

**UCLA** Los Angeles, CA — Sep 2021 - Present

Graduate Student Researcher

- Lead developer of CHEERSafe, a parallelized Python library (11K lines, 500+ tests) estimating hurricane wind and flood losses for 1.15M single-family homes in 45 North Carolina counties; damage-library generation runs 65× faster on 112 cores. [🔗]
- Built a linear-programming calibration (Gurobi) that assigns annual probabilities to 932 hurricane scenarios to match 5,400 ASCE-7 wind targets and FEMA flood-zone targets for the NSF CoPe-CHEER project.
- Implemented North Carolina's AI-driven building-inventory pipeline (5.65M buildings from FEMA, Microsoft, NSI, and parcel data) with a ViT roof-shape classifier (F1 0.92) and gradient-boosting imputation cutting structure-value error by 37%.
- Co-first author of RAPID-A (*ASCE OPEN*, 2025), an uncertainty-aware ensemble of deep segmentation models that assesses post-earthquake building damage from satellite images of the 2023 Türkiye earthquakes and generalizes to unseen cities.
- Co-developed an XGBoost claim-occurrence model (736K building–hurricane observations, 4 hurricanes) that cut total claim-count error to 5.6% from 42.1% with logistic regression on held-out data; 2nd author, *Natural Hazards* (2026).
- Developed transformer-based NLP and ML techniques for PII removal and text classification of damage descriptions for the North Carolina Insurance Underwriting Association (NCIUA).
- Rebuilt ground-motion record selection in JAX as a GPU-resident greedy search: 2,000×+ faster than the legacy OpenCL tool [] (122 s → 0.06 s for 200 records) with equal or better mean-spectrum fit.

## PROJECTS (SELECTED)

### Open-source and graduate-level

- **Overleaf Web MCP** (open source; TypeScript, Node.js): Built and shipped a Model Context Protocol (MCP) server with 31 tools that lets AI assistants (Claude Code, Codex, Cursor) read, edit, compile, and review Overleaf projects through the user's own session; 2.3K+ npm downloads in three months. Engineered safeguards for agent-driven edits: revision-checked writes, tracked changes, and plan-then-confirm folder sync. []
- **Proper Scoring Rules to Reduce LLM Overconfidence** (Python, LLM evaluation): Demonstrated that stating the logarithmic scoring rule in the prompt curbs DeepSeek-V4-Flash's overconfidence: +0.54 nats in log score on MMLU and +1.49 on MMLU-Pro (1,000 questions each) at negligible accuracy cost. []
- **Causal Reasoning Agent** (team of 4; Python): Co-developed an LLM-agnostic agent framework with Kripke epistemic reasoning; built its Mastermind and 2048 environments, evaluation runners, and a decision-trace dashboard. []
- Built an efficient ML pipeline that optimizes XGBoost hyperparameters via experimental design techniques, outperforming grid search in predicting wildfire damage to buildings. — Advisor: Professor Hongquan Xu
- Implemented Deep Q-learning with experience replay (DQN) and Deep Deterministic Policy Gradient (DDPG) reinforcement learning algorithms for Racing Car and Double Pendulum problems. — Advisor: Professor Jonathan Kao
- Evaluated performance of CRNN, LSTM, conditional GANs, and CNN models on time series data using TensorFlow, highlighting robust deep learning methods for signal processing. — Advisor: Professor Jonathan Kao []
- A tutorial on the doubly robust targeted maximum likelihood (TMLE) estimation method and its implementation in R, merging causal inference with ML principles for improved estimation. — Advisor: Professor Chad Hazlett

## TECHNICAL SKILLS

**Programming Languages**: Python, C++, R, TypeScript, SQL, Bash, MATLAB
**ML & DL Frameworks**: PyTorch, JAX, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, spaCy
**LLMs & Agents**: LangChain, LlamaIndex, RAG, MCP, Google Cloud Vertex AI
**Tools**: Git, GitHub, CI/CD, Docker, OpenCL, Gurobi

# PUBLICATIONS AND PRESENTATIONS

## Articles

[1] N. O. Tackie-Otoo, **M. Askari**, P. Hadinata, R. A. Davidson, E. Taciroglu, and G. Hardy, "Hurricane wind loss modeling using insurance claims data," *Natural Hazards*, vol. 122, no. 7, p. 300, 2026. doi: 10.1007/s11069-026-08059-z.

[2] **M. Askari**, M. H. Soleimani-Babakamali, C. Williams, R. Davidson, and E. Taciroglu, *CHEERSafe: A Python Library to Assess Hurricane Damage and Loss*, Manuscript in preparation, 2026.

[3] M. H. Soleimani-Babakamali, **M. Askari**, M. A. Heravi, R. Sisman, N. Attarchian, A. Askan, R. Soleimani, and E. Taciroglu, "Deep Ensemble Learning for Rapid Large-Scale Postearthquake Damage Assessment: Application to Satellite Images from the 2023 Türkiye Earthquakes," *ASCE OPEN: Multidisciplinary Journal of Civil Engineering*, vol. 3, no. 1, p. 04 025 003, 2025. doi: 10.1061/AOMJAH.AOENG-0043.

[4] S.-Z. Chen, X.-Y. Wang, X.-R. Ma, T. Wang, **M. Askari**, and E. Taciroglu, "Transforming Regression to Classification for Enhancing Extrapolation Capacity of Data-driven Structural Behavior Prediction," *Expert Systems with Applications*, vol. 277, p. 127 302, 2025, issn: 0957-4174. doi: https://doi.org/10.1016/j.eswa.2025.127302.

[5] R. Soleimani-Babakamali, M. H. Soleimani-Babakamali, M. A. Heravi, **M. Askari**, O. Avci, and E. Taciroglu, "Transferring Damage Detection Knowledge Across Rotating Machines and Framed Structures: Harnessing Domain Adaptation and Contrastive Learning," *Mechanical Systems and Signal Processing*, vol. 221, p. 111 743, 2024. doi: https://doi.org/10.1016/j.ymssp.2024.111743.

[6] M. Shokrabadi, Y. Bozorgnia, H. V. Burton, J. W. Baker, and **M. Askari**, "An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra," University of California, Los Angeles, Tech. Rep., 2022. doi: https://doi.org/10.34948/N3K01N.

[7] **M. Askari** and M. Mahsuli, "Modeling of Epistemic Uncertainties in Seismic Hazard Analysis Based on Reliability Methods," Center for Infrastructure Sustainability and Resilience Research, Sharif University of Tech., Tech. Rep., 2020.

## Presentations

| | | |
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| May 2025 | EMI Conference | CHEERSafe: A Modular Object-Oriented Framework for Regional Hurricane Risk Assessment |
| May 2024 | PMC/EMI Conference | An End-To-End Framework for Building Inventory Generation Through Public Data Fusion and Artificial Intelligence |
| Feb 2024 | NHERI Computational Symposium | Deep Ensemble Learning for Rapid Large-Scale Post-Earthquake Damage Assessment—Application to 2023 Turkiye Earthquakes Satellite Images (*poster*) |
| Feb 2022 | ASCE Lifelines Conference | An Efficient Computational Platform for Selecting and Scaling Ground Motion Records while Considering Multiple Target Spectra (*poster*) |

# CERTIFICATIONS

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| Jul 2024 | TACC | AI Computational Academy & Hackathon |
| Jan 2024 | Coursera | AI for Good Specialization [🔗] |
| Nov 2023 | NVIDIA | Building Transformer-Based NLP Applications [🔗] |
| May 2023 | Coursera | Deep Learning Specialization [🔗] |
| Feb 2021 | Coursera | Machine Learning [🔗] |

# COMMUNITY ENGAGEMENT & VOLUNTEER EXPERIENCE

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|---|---|---|
| Sep 2024-present | Elsevier Engineering Structures<br>ASCE Open<br>ASCE I3CE | Journal article reviewer related to ML and AI applications |
| 2024–2025 | LA Hacks | Expert judge for the LA Hacks hackathon |
| Feb 2024-present | NSF | Code and documentation management of CoPe-CHEER NSF project |
| Sep 2022-July 2023 | UCLA | Board member of the Iranian Graduate and Professional Association, IGPA |